

Advanced approximate and symbolic computing

Tamás Vinkó

University of Szeged



Orthogonal transformations II

Elementary rotation matrices – Givens

- Wallace Givens, during the '50s worked at Argonne National Library and introduced the usage of rotation matrices in numerical mathematics.

Definition 2.1 For any $1 \leq p < q \leq n$ and $0 \leq \theta \leq 2\pi$ the matrix of form

$$\mathbf{G}(p, q, \theta) = \mathbf{I} + (\cos \theta - 1)(\mathbf{e}_p \mathbf{e}_p^T + \mathbf{e}_q \mathbf{e}_q^T) + \sin \theta (\mathbf{e}_p \mathbf{e}_q^T - \mathbf{e}_q \mathbf{e}_p^T) \quad (1)$$

is called elementary rotation matrix.

(Also called: Givens-matrices.)

This is a rotation on the (p, q) plain by θ radians.

How do they look like?

$$\mathbf{G}(p, q, \theta) = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \cos \theta & \sin \theta & & \\ & & & -\sin \theta & \cos \theta & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \begin{matrix} \leftarrow (p) \\ \leftarrow (q) \end{matrix}$$

$\uparrow$
 (p)

$\uparrow$
 (q)

and all non-filled elements are the same as the corresponding elements of the identity matrix $\mathbf{I}$.

Special case

The most simple case: 2×2 elementary rotation matrix, which has the form

$$\mathbf{G}(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Given a 2×2 matrix

$$\mathbf{A} = \begin{pmatrix} a_{pp} & a_{pq} \\ a_{qp} & a_{qq} \end{pmatrix}$$

we are looking for the case when the matrix $\mathbf{G}(\theta)\mathbf{A}$ has zero at its component $(2, 1)$, i.e.,

$$a_{qp} \cos \theta - a_{pp} \sin \theta = 0.$$

- We do not need to calculate the value of θ
- we use $\cos \theta$ and $\sin \theta$ directly and do the calculations based on the followings.
- We know from trigonometry that $\cos^2 \theta + \sin^2 \theta = 1$,
- this needs to hold, hence let

$$d := \sqrt{a_{pp}^2 + a_{qp}^2},$$

and

$$\cos \theta := \frac{a_{pp}}{d} \quad \text{and} \quad \sin \theta := \frac{a_{qp}}{d},$$

assuming that $d \neq 0$.

- If $d = 0$, then $\theta = 0$, meaning that $\mathbf{G}(\theta)$ is the identity matrix.

This can be extended to the case of $n \times n$ elementary rotation matrices. The most important properties of these matrices are summarized in the following lemma.

Lemma 2.2 *For any rotation matrix $\mathbf{G}(p, q, \theta)$ and matrix $\mathbf{A}$ the followings hold:*

(i) *the matrix $\mathbf{G}(p, q, \theta)$ is orthogonal.*

(ii) *the matrix rowvectors $\mathbf{a}'_1, \mathbf{a}'_2, \dots, \mathbf{a}'_n$ of the matrix $\mathbf{A}' = \mathbf{G}(p, q, \theta)\mathbf{A}$ for any $1 \leq m \leq n$ indices can be calculated from the row vectors of matrix $\mathbf{A}$ using the formulas:*

$$\mathbf{a}'_m = \begin{cases} \mathbf{a}_m & \text{if } m \neq p \text{ and } m \neq q \\ \cos \theta \mathbf{a}_p + \sin \theta \mathbf{a}_q & \text{if } m = p \\ -\sin \theta \mathbf{a}_p + \cos \theta \mathbf{a}_q & \text{if } m = q \end{cases}$$

Complexity

Lemma 2.3 *For any rotation matrix $\mathbf{G}(p, q, \theta)$, matrix $\mathbf{A}$ and vector $\mathbf{x}$ we need to use*

- *for calculating the $\mathbf{G}(p, q, \theta)\mathbf{x}$ transformed vector $O(1)$ operations,*
- *and for the $\mathbf{G}(p, q, \theta)\mathbf{A}$ transformed matrix $O(n)$ operations.*

Proof. Based on the definition (1) and with the usage of Lemma 2.2 this can be proved. ■

Partitioned form

Lemma 2.4 *Let $1 \leq k < n$ be arbitrary, the matrix $\mathbf{G} \in \mathcal{R}^{k \times k}$ is a rotation matrix. Then the matrices*

$$\mathbf{T} \stackrel{\text{def}}{=} \left(\begin{array}{c|c} \mathbf{I} & \mathbf{O} \\ \hline \mathbf{O}^T & \mathbf{G} \end{array} \right) \in \mathcal{R}^{n \times n} \quad \text{and} \quad \mathbf{W} \stackrel{\text{def}}{=} \left(\begin{array}{c|c} \mathbf{G} & \mathbf{O} \\ \hline \mathbf{O}^T & \mathbf{I} \end{array} \right) \in \mathcal{R}^{n \times n}$$

will also be rotation matrices. (Here, $\mathbf{I}$ and $\mathbf{O}$ are standing for the unity and zero matrices.)

Orthogonal triangularization by Given method

Theorem 2.5 *Any matrix $\mathbf{A} \in \mathcal{R}^{n \times n}$ can be factorized by multiplication of elementary rotation matrices. More precisely, we can define finite many elementary matrices $\mathbf{G}(p, q, \theta_{p,q})$:*

$$\begin{aligned} &\mathbf{G}(1, 2, \theta_{1,2}), \mathbf{G}(1, 3, \theta_{1,3}), \dots, \mathbf{G}(1, n, \theta_{1,n}), \\ &\quad \mathbf{G}(2, 3, \theta_{2,3}), \dots, \mathbf{G}(2, n, \theta_{2,n}), \\ &\quad \dots, \mathbf{G}(n-1, n, \theta_{n-1,n}) \end{aligned}$$

in such a way that the product

$$\mathbf{G}(n-1, n, \theta_{n-1,n}) \dots \mathbf{G}(1, 3, \theta_{1,3}) \mathbf{G}(1, 2, \theta_{1,2}) \mathbf{A}$$

will be upper triangular.

Example - construction of a Givens matrix

Let's take the following matrix in which we want to make element $(1,3)$ into zero.

$$\mathbf{A} = \begin{pmatrix} 12 & -51 & 4 \\ 6 & 167 & -68 \\ -4 & 24 & -41 \end{pmatrix}$$

We have $q = 3$ and $p = 1$, where the appropriate Givens matrix has the form

$$\mathbf{G} = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix},$$

where $\cos \theta = a_{pp}/d = a_{11}/d$ and $\sin \theta = a_{qp}/d = a_{31}/d$, moreover

$$d = \sqrt{a_{qp}^2 + a_{pp}^2} = \sqrt{a_{31}^2 + a_{11}^2} = \sqrt{12^2 + (-4)^2} \approx 12.6491.$$

Hence, $\cos \theta \approx 0.9487$ and $\sin \theta \approx -0.3162$.

Complexity

As we have seen earlier, a rotation can be done with using $O(n)$ operations, hence

Corollary 2.6 *Any matrix $\mathbf{A} \in \mathcal{R}^{n \times n}$ can be factorized in the form $\mathbf{A} = \mathbf{QR}$ using $O(n^3)$ operations.*

Further notes on the QR decomposition

QR decomposition is not unique

Hint: consider how we obtain the element a_{11} .

Since the first column of matrix $\mathbf{R}$ is all zero except its first element, we have

$$a_{11} = q_{11}r_{11}.$$

Now, there are two possibilities:

- if $a_{11} > 0$ then either both q_{11} and r_{11} are positive or both are negative;
- if $a_{11} < 0$ then the signs of q_{11} and r_{11} need to be different.

QR factorization for non-square matrix

As we have already mentioned, the **QR** factorization can be done for *any* matrix.

For us, the important case is when matrix **A** has more rows than columns. The following theorem is regarding this case.

Theorem 2.7 *For any $m \times n$ -dimensional matrix **A**, where $m \geq n$, the factorization can be done in the form of*

$$\mathbf{A} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{O} \end{pmatrix} \begin{matrix} n \text{ rows} \\ m - n \text{ rows} \end{matrix}$$

*where **Q** is an $m \times m$ -dimensional orthogonal matrix, and **R** is an $n \times n$ -dimensional upper triangular matrix.*

Applications of QR factorization

- (Computing the matrix determinant)
- (Inverting the matrix)
- Solving system of linear equations
 - there is a special case to be discussed here
- Most importantly: eigenvalues